



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO 3 – Assessment 1

Experiments

Experiment 1

Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Aim:

To create a simple cloud-based Library Book Reservation System for SIMATS Library using a Cloud Service Provider (SaaS).

Procedure:

1. Log in to a cloud service provider (e.g., Zoho Creator).

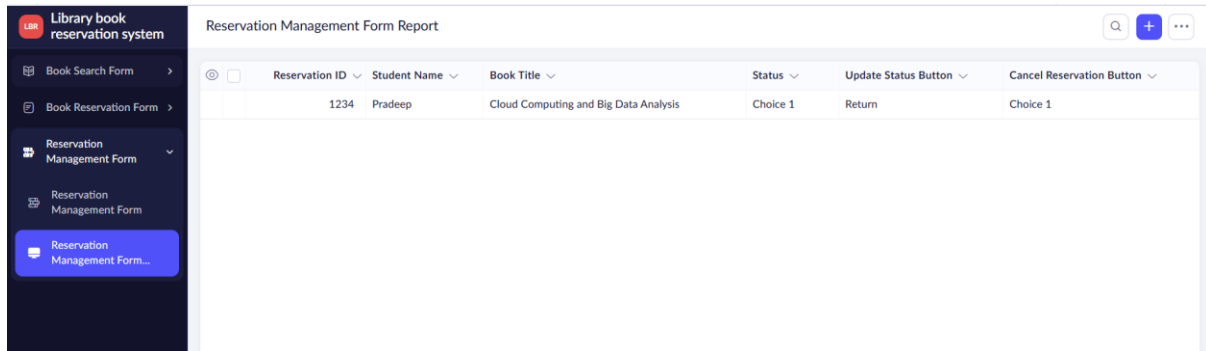
2. Create a new application named Library Book Reservation System.
3. Create a form by adding the required library book details such as Title, Author, Publication, Year, Edition, Number of Copies, Rack Number, and Status.
4. Save the application.
5. Enter sample library book details.
6. Test the application by adding, updating, and viewing records.

Snapchat:

The screenshot displays the 'Library book reservation system' interface. On the left is a dark sidebar with a menu containing 'Book Search Form', 'Book Reservation Form', 'Reservation Management Form' (selected), and 'Reservation Management Form...'. The main area is titled 'Reservation Management Form' and contains the following fields:

- Reservation ID:** A text input field containing '1234'.
- Student Name:** Two text input fields. The first, labeled 'First Name', contains 'Pradeep'. The second, labeled 'Last Name', is empty.
- Book Title:** A text input field containing 'Cloud Computing and Big Data Analysis'.
- Status:** A dropdown menu showing 'Choice 1' with a close (X) and expand (V) icon.
- Update Status Button:** A text input field containing 'Return'.
- Cancel Reservation Button:** A group of three radio buttons labeled 'Choice 1', 'Choice 2', and 'Choice 3'. The 'Choice 1' radio button is selected.

A blue 'Submit' button is located at the bottom center of the form.



The screenshot displays a web application for a library book reservation system. On the left is a dark sidebar with navigation options: 'Book Search Form', 'Book Reservation Form', and 'Reservation Management Form' (which is expanded to show 'Reservation Management Form' and 'Reservation Management Form...'). The main area is titled 'Reservation Management Form Report' and contains a table with the following data:

Reservation ID	Student Name	Book Title	Status	Update Status Button	Cancel Reservation Button
1234	Pradeep	Cloud Computing and Big Data Analysis	Choice 1	Return	Choice 1

Result:

Thus, a cloud-based Library Book Reservation System was successfully created using the SaaS model.

Experiment 2

Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org, over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

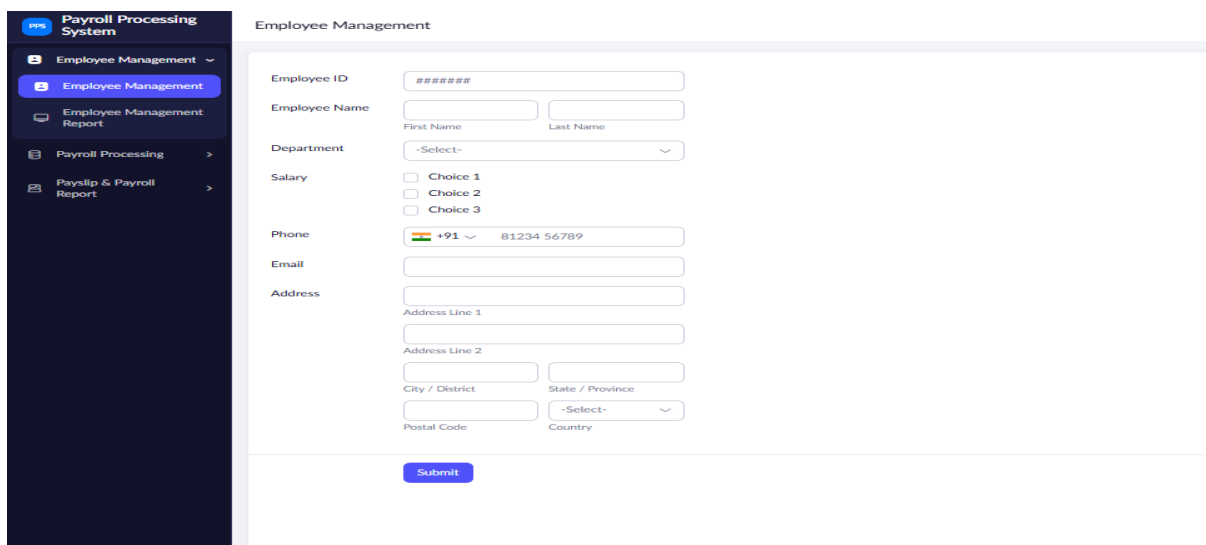
Aim:

To create a cloud-based Payroll Processing System using a Cloud Service Provider to demonstrate SaaS.

Procedure:

1. Open the cloud platform (e.g., Zoho Creator) and sign in with your account.
2. Create a new application for the Library Book Reservation System.
3. Create a form to store library book information and reservation details.
4. Add the required fields such as Title, Author, Publication, Year, Edition, Number of Copies, Rack Number, and Status (Taken/Returned).
5. Save the form and deploy the application.
6. Enter sample book records into the application.
7. Verify that the records can be added, updated, and viewed successfully.

Snapchat:



The screenshot displays the 'Employee Management' form within the 'Payroll Processing System'. The interface features a dark blue sidebar on the left with navigation links: 'Employee Management' (selected), 'Employee Management Report', 'Payroll Processing', and 'Payslip & Payroll Report'. The main content area is titled 'Employee Management' and contains the following fields:

- Employee ID:** A text field with a placeholder '#####'.
- Employee Name:** Two text fields for 'First Name' and 'Last Name'.
- Department:** A dropdown menu with the option '-Select-'.
- Salary:** Three radio button options: 'Choice 1', 'Choice 2', and 'Choice 3'.
- Phone:** A text field with a country code dropdown set to '+91' and a placeholder '81234 56789'.
- Email:** A text field.
- Address:** Three text fields for 'Address Line 1', 'Address Line 2', and 'City / District'.
- State / Province:** A dropdown menu with the option '-Select-'.
- Postal Code:** A text field.
- Country:** A dropdown menu with the option '-Select-'.

A blue 'Submit' button is located at the bottom of the form.

Payroll Processing System

Employee Management

Employee Management

Employee Management Report

Payroll Processing

Payslip & Payroll Report

Employee Management Report

Employee ID

Employee Name

Department

Salary

Phone

Email

Address

456

Prithiv

Choice 2

Choice 1

+919585994260

prithiv@gmail.com

7/41, Senggaipatti, Buttalur, Thanjavur, Tamil Nadu, 234567, India

Result:

Thus, a cloud-based Library Book Reservation System was successfully created using the SaaS model.

Experiment 3

Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Aim:

To install a Type-2 Hypervisor, create and configure a virtual machine with 1 CPU, 2 GB RAM, and 15 GB storage.

Procedure:

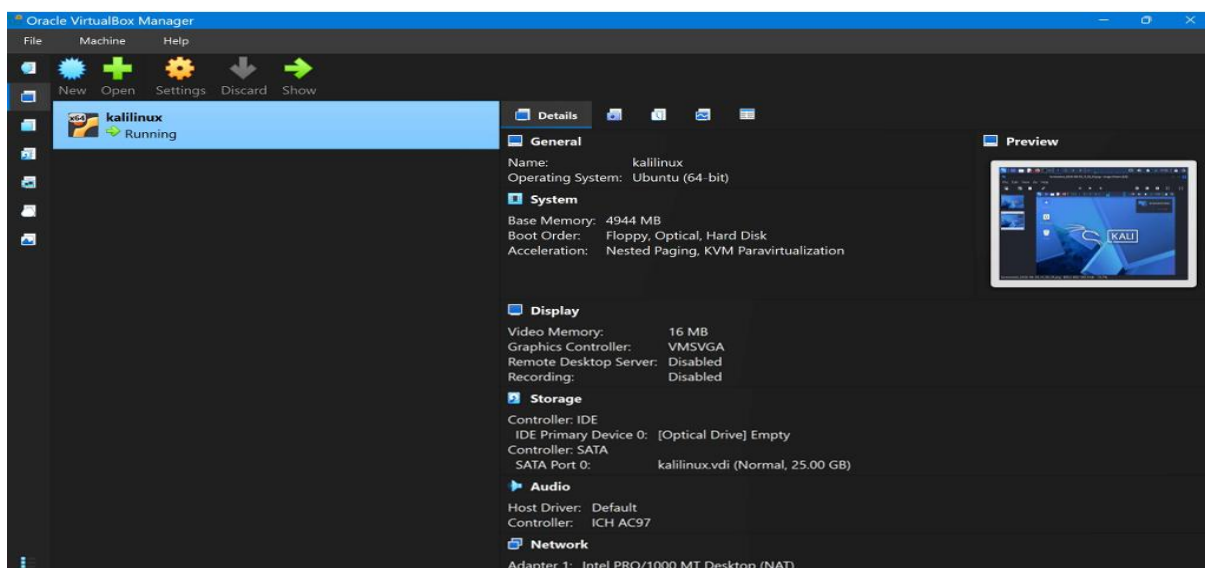
1. Install a Type-2 Hypervisor (Oracle VirtualBox or VMware Workstation).
2. Open the virtualization software.
3. Click New to create a virtual machine.
4. Enter the VM name and select the operating system (Windows/Linux).
5. Allocate: CPU: 1 Processor

RAM: 2 GB (2048 MB)

Storage: 15 GB Virtual Disk

6. Attach the operating system ISO file.
7. Start the virtual machine.
8. Install the operating system.
9. Verify that the VM boots successfully.

Snapchat:





Result:

Thus, a virtual machine with 1 CPU, 2 GB RAM, and 15 GB storage was successfully created and configured using a Type-2 Hypervisor.

Experiment 4

Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

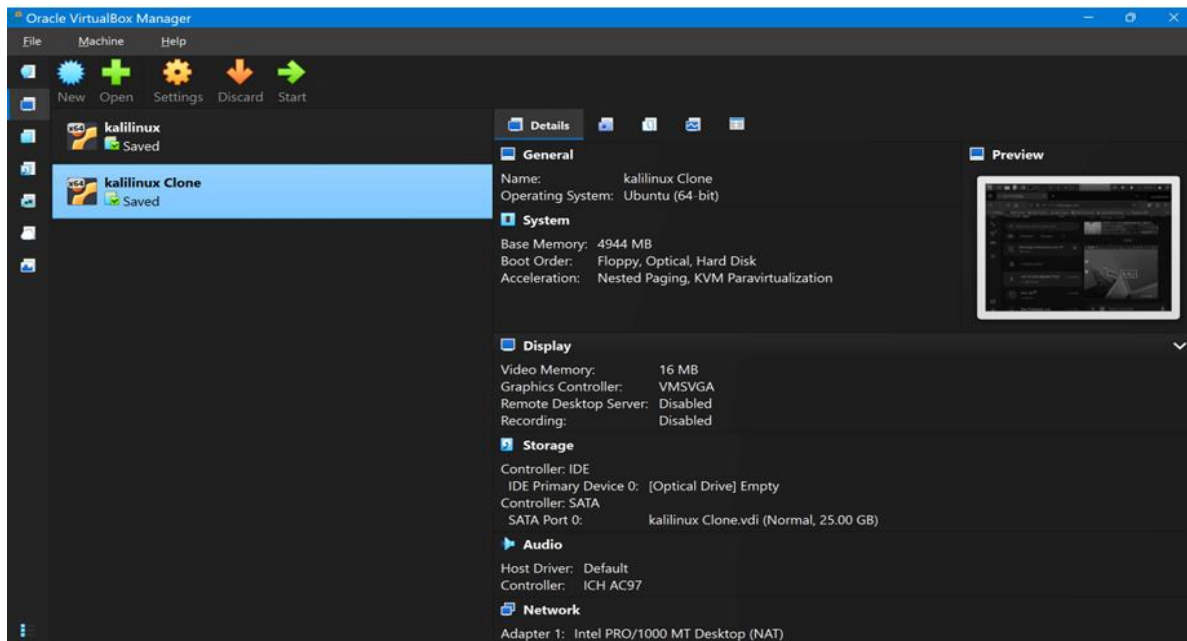
Aim:

To create a clone of an existing virtual machine and verify its functionality using a Type-2 Hypervisor.

Procedure:

1. Open the Type-2 Hypervisor (VirtualBox/VMware).
2. Select the existing virtual machine.
3. Right-click the VM and choose Clone.
4. Enter a name for the cloned virtual machine.
5. Select Full Clone or Linked Clone.
6. Complete the cloning process.
7. Start the cloned virtual machine.
8. Verify that the operating system loads correctly.
9. Compare the cloned VM with the original VM to ensure successful cloning.

Snapchat:



Result:

Thus, the virtual machine was successfully cloned and the cloned VM was tested by loading the previous version, demonstrating virtualization using a Type-2 Hypervisor.